Ad Hoc Transactions through the Looking Glass: An Empirical Study of Application-Level Transactions in Web Applications

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Many transactions in web applications are constructed ad hoc in the application code. For example, developers might explicitly use locking primitives or validation procedures to coordinate critical code fragments. We refer to database operations coordinated by application code as *ad hoc transactions*. Until now, little is known about them. This paper presents the first comprehensive study on ad hoc transactions. By studying 91 ad hoc transactions among 8 popular open-source web applications, we found that (i) every studied application uses ad hoc transactions (up to 16 per application), 71 of which play critical roles; (ii) compared with database transactions, concurrency control of ad hoc transactions is much more flexible; (iii) ad hoc transactions are error-prone—53 of them have correctness issues, and 33 of them are confirmed by developers; and (iv) ad hoc transactions have the potential for improving performance in contentious workloads by utilizing application semantics such as access patterns. Based on these findings, we discuss the implications of ad hoc transactions to the database research community.

CCS Concepts: • **Information systems** → **Database transaction processing**; **Web applications**.

Additional Key Words and Phrases: ad hoc transactions

1 INTRODUCTION

Today, web applications often use database systems to persist large amounts of data, necessitating the coordination of concurrent database operations for correctness. One common approach is using database transactions. Transactions isolate concurrent database operations by encapsulating them into individual units of work. Another widely adopted approach is using the invariant validation APIs provided by object-relational mapping (ORM) frameworks (e.g., the validates keyword from Active Record [86]). With such APIs, developers explicitly specify invariants, such as the uniqueness of column values, in the application code and the ORM frameworks report errors on invariant violations. So far, much work has been done to investigate and improve these two approaches [7, 8, 21, 26, 32, 54, 55, 67, 85, 103, 106].

However, besides these approaches, application developers are also accustomed to coordinating critical database operations ad hoc. Specifically, developers might explicitly use locking primitives and validation procedures to implement concurrency control (CC), e.g., optimistic concurrency control (OCC), amid the application code to coordinate critical database operations. We refer to such ad hoc coordination of database operations as *ad hoc transactions*. Developers' comments suggest that they implement ad hoc transactions for flexibility or efficiency [20].

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